Convergence Analysis of Stochastic Variance Reduced Gradient (SVRG) for Non-Convex Smooth Optimization

Algorithm Name

Stochastic Variance Reduced Gradient (SVRG) for non-convex finite-sum minimization.

We consider the finite-sum problem

$$\min_{x \in \mathbb{R}^d} f(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x),$$

where each f_i is smooth but f may be non-convex.

Mathematical Formulation

Double-loop structure. SVRG maintains a *snapshot* point $\tilde{x}^s$ at the start of each outer iteration s. The full gradient is computed only at the snapshot:

$$\tilde{g}^s = \nabla f(\tilde{x}^s) = \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(\tilde{x}^s).$$

Inner loop (variance-reduced estimator). For $t = 0, 1, \ldots, m-1$, draw an index i_t uniformly at random from $\{1, \ldots, n\}$ and form the estimator

$$v_t = \nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x}^s) + \tilde{g}^s.$$

The update rule is

$$x_{t+1} = x_t - \eta\, v_t,$$

where $\eta > 0$ is a fixed step size. (Note: the problem statement writes the estimator as $\nabla f(x_t) - \nabla f(x_{t-1}) + \text{full\_grad}$; the standard SVRG form referencing the snapshot $\tilde{x}^s$ is used here, which is the mathematically well-posed convention.)

Snapshot update. After m inner steps, the new snapshot is taken as

$$\tilde{x}^{s+1} = x_m \quad \text{(Option I)}.$$

This is the convention actually used in the convergence analysis below; the telescoping argument relies on $f(\tilde{x}^{s+1}) = f(x_m)$.

Hyperparameters.

- Step size $\eta > 0$.

- Inner loop length m.
- Number of outer epochs S.
- Lipschitz constant L of the gradients.

Unbiasedness. Conditioned on x_t (and the snapshot $\tilde{x}^s$),

$$\mathbb{E}_{i_t}[v_t] = \mathbb{E}_{i_t}[\nabla f_{i_t}(x_t)] - \mathbb{E}_{i_t}[\nabla f_{i_t}(\tilde{x}^s)] + \tilde{g}^s = \nabla f(x_t) - \nabla f(\tilde{x}^s) + \nabla f(\tilde{x}^s) = \nabla f(x_t).$$

Pseudocode

```
Input: initial point x_tilde^0, step size eta, inner length m, epochs S
for s = 0, 1, ..., S-1 do
    g_tilde = (1/n) * sum_i grad f_i(x_tilde^s)   # full gradient at snapshot
    x_0 = x_tilde^s
    for t = 0, 1, ..., m-1 do
        sample i_t uniformly from {1,...,n}
        v_t = grad f_{i_t}(x_t) - grad f_{i_t}(x_tilde^s) + g_tilde
        x_{t+1} = x_t - eta * v_t
    end for
    # Option I (used in the analysis):
    x_tilde^{s+1} = x_m
end for
Output: x_a chosen uniformly from all inner iterates {x_t^s : 0<=t<=m-1, 0<=s<=S-1}
```

Dry Run Example

Let $f(w) = (w-3)^2$ (single component, so $n = 1$, $f_1 = f$), $w_0 = 0$, $\eta = 0.1$. The gradient is $\nabla f(w) = 2(w-3)$.

Since $n = 1$, the variance-reduced estimator reduces exactly to the full gradient:

$$v_t = \nabla f_1(x_t) - \nabla f_1(\tilde{x}) + \nabla f(\tilde{x}) = \nabla f(x_t).$$

Thus the inner loop is exactly gradient descent.

Iteration 1:

$$\begin{aligned}
\nabla f(w_0) &= 2(0-3) = -6, \\
w_1 &= w_0 - \eta \nabla f(w_0) = 0 - 0.1(-6) = 0.6, \\
f(w_1) &= (0.6-3)^2 = (-2.4)^2 = 5.76.
\end{aligned}$$

Iteration 2:

$$\begin{aligned}
\nabla f(w_1) &= 2(0.6-3) = -4.8, \\
w_2 &= 0.6 - 0.1(-4.8) = 0.6 + 0.48 = 1.08, \\
f(w_2) &= (1.08-3)^2 = (-1.92)^2 = 3.6864.
\end{aligned}$$

Iteration 3:

$$\begin{aligned}\nabla f(w_2) &= 2(1.08-3) = -3.84,\\ w_3 &= 1.08 - 0.1(-3.84) = 1.08 + 0.384 = 1.464,\\ f(w_3) &= (1.464-3)^2 = (-1.536)^2 = 2.359296.\end{aligned}$$

Objective decreases $9 \to 5.76 \to 3.6864 \to 2.3593$, converging toward the minimizer $w^* = 3$.

Assumptions

Assumption 1 (L-smoothness)**.** *Each f_i is L-smooth: for all x, y,*

$$\|\nabla f_i(x) - \nabla f_i(y)\| \le L\|x-y\|.$$

Consequently f is L-smooth and the descent lemma holds:

$$f(y) \le f(x) + \langle \nabla f(x), y-x\rangle + \frac{L}{2}\|y-x\|^2.$$

Assumption 2 (Lower boundedness)**.** *f is bounded below: $f^\star = \inf_x f(x) > -\infty$.*

Assumption 3 (Unbiased sampling)**.** *Indices i_t are drawn i.i.d. uniformly, so $\mathbb{E}_{i_t}[\nabla f_{i_t}(x)] = \nabla f(x)$.*

Main Convergence Theorem

Theorem 1 (Non-convex SVRG convergence)**.** *Under Assumptions 1–3, run SVRG with the Option I snapshot rule $\tilde{x}^{s+1} = x_m$. Let the output x_a be chosen uniformly at random from the set of inner iterates $\{x_t^s : 0 \le t \le m-1,\ 0 \le s \le S-1\}$ (total $T = mS$). Choose inner length $m = \lfloor n^{1/2} \rfloor$ and step size $\eta = \mu_0/(L\, n^{1/3})$ for a sufficiently small constant $\mu_0 \in (0,1]$. Then there is a constant $\Gamma \ge \eta/2 > 0$ (depending only on μ_0, L, n) such that*

$$\mathbb{E}\big[\|\nabla f(x_a)\|^2\big] \le \frac{f(\tilde{x}^0) - f^\star}{T\,\Gamma}.$$

In particular, with the constants above,

$$\mathbb{E}\big[\|\nabla f(x_a)\|^2\big] \le \frac{2L\, n^{1/3}\,(f(\tilde{x}^0) - f^\star)}{\mu_0\, T} = O\!\left(\frac{Ln^{1/3}\,(f(\tilde{x}^0) - f^\star)}{T}\right).$$

The rate is $O(n^{1/3}/T)$; the dependence on n does not *vanish and the bound is* not *n-independent.*

Theoretical Properties

- **Variance reduction.** The key property is that $\mathbb{E}\|v_t - \nabla f(x_t)\|^2$ shrinks as $x_t, \tilde{x}^s \to x^\star$, unlike plain SGD whose variance is a fixed σ^2.
- **Convergence to stationarity.** For non-convex f we measure progress by $\mathbb{E}\|\nabla f(x_a)\|^2$, the standard stationarity criterion.

- **Complexity.** Total gradient computations to reach $\mathbb{E}\|\nabla f(x_a)\|^2 \le \epsilon$ is $O(n + n^{2/3}/\epsilon)$ (see Rate Derivation), improving over GD ($O(n/\epsilon)$) and SGD ($O(1/\epsilon^2)$).
- **Step-size sensitivity.** $\eta = O(1/(Ln^{1/3}))$ is required; too large η breaks the Lyapunov contraction.
- **Stability.** The double loop periodically recomputes the full gradient, anchoring the estimator and bounding accumulated error.

Discussion

Compared to SGD, whose non-convex stationarity rate is $O(1/\sqrt{T})$, SVRG attains the faster $O(n^{1/3}/T)$ rate in $\mathbb{E}\|\nabla f\|^2$ thanks to variance reduction. The $n^{1/3}$ factor reflects the problem size; the resulting oracle complexity $O(n + n^{2/3}/\epsilon)$ improves on full-batch GD's $O(n/\epsilon)$. The trade-off is the periodic full-gradient cost of n component gradients per epoch.

Preliminaries (Definitions and Lemmas)

Throughout an epoch we drop the superscript s and write $\tilde{x} = \tilde{x}^s$. $\mathbb{E}_t[\cdot]$ denotes expectation over i_t conditioned on x_t.

Lemma 1 (Unbiasedness). $\mathbb{E}_t[v_t] = \nabla f(x_t)$.

Proof. By Assumption 3, $\mathbb{E}_t[\nabla f_{i_t}(x_t)] = \nabla f(x_t)$ and $\mathbb{E}_t[\nabla f_{i_t}(\tilde{x})] = \nabla f(\tilde{x})$. Hence

$$\mathbb{E}_t[v_t] = \nabla f(x_t) - \nabla f(\tilde{x}) + \nabla f(\tilde{x}) = \nabla f(x_t). \qquad \text{[Step 1]}$$

□

Lemma 2 (Second moment bound of the estimator). *Under Assumption 1,*

$$\mathbb{E}_t\|v_t\|^2 \le 2\|\nabla f(x_t)\|^2 + 2L^2\, \mathbb{E}_t\|x_t - \tilde{x}\|^2.$$

Proof. Write

$$v_t = \nabla f(x_t) + \underbrace{\left(\nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x}) - (\nabla f(x_t) - \nabla f(\tilde{x}))\right)}_{=:\,\xi_t}. \qquad \text{[Step 2]}$$

Indeed $\nabla f(\tilde{x}) = \tilde{g}^s$, so $v_t = \nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x}) + \nabla f(\tilde{x})$ rearranges to the above. Using $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$:

$$\mathbb{E}_t\|v_t\|^2 \le 2\|\nabla f(x_t)\|^2 + 2\,\mathbb{E}_t\|\xi_t\|^2. \qquad \text{[Step 3]}$$

For ξ_t, note it is a centered random variable; using $\mathbb{E}\|X - \mathbb{E}X\|^2 \le \mathbb{E}\|X\|^2$ with $X = \nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x})$:

$$\mathbb{E}_t\|\xi_t\|^2 \le \mathbb{E}_t\|\nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x})\|^2 \le L^2\|x_t - \tilde{x}\|^2, \qquad \text{[Step 4]}$$

where the last inequality is L-smoothness of each f_i. Combining [Step 3] and [Step 4] gives the claim. □

Lemma 3 (Iterate drift bound). *For the inner iterates with $x_0 = \tilde{x}$,*

$$\mathbb{E}\|x_{t+1} - \tilde{x}\|^2 \le (1+\beta)\,\mathbb{E}\|x_t - \tilde{x}\|^2 + \left(1 + \tfrac{1}{\beta}\right)\eta^2\,\mathbb{E}\|v_t\|^2,$$

for any $\beta > 0$.

Proof. From $x_{t+1} = x_t - \eta v_t$,

$$\|x_{t+1} - \tilde{x}\|^2 = \|x_t - \tilde{x}\|^2 - 2\eta\langle v_t, x_t - \tilde{x}\rangle + \eta^2\|v_t\|^2. \qquad \text{[Step 5]}$$

Apply Young's inequality $-2\langle a, b\rangle \le \beta\|b\|^2 + \frac{1}{\beta}\|a\|^2$ with $a = \eta v_t$, $b = x_t - \tilde{x}$:

$$-2\eta\langle v_t, x_t - \tilde{x}\rangle \le \beta\|x_t - \tilde{x}\|^2 + \tfrac{1}{\beta}\eta^2\|v_t\|^2. \qquad \text{[Step 6]}$$

Substituting into [Step 5] and taking expectations gives the result. □

Main Proof (Step-by-Step Derivation)

Step A: Descent lemma applied to the update. By Assumption 1 (descent lemma) with $y = x_{t+1}$, $x = x_t$:

$$f(x_{t+1}) \le f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t\rangle + \frac{L}{2}\|x_{t+1} - x_t\|^2. \qquad \text{[Step 7]}$$

Substituting $x_{t+1} - x_t = -\eta v_t$:

$$f(x_{t+1}) \le f(x_t) - \eta\langle \nabla f(x_t), v_t\rangle + \frac{L\eta^2}{2}\|v_t\|^2. \qquad \text{[Step 8]}$$

Taking conditional expectation $\mathbb{E}_t$ and using Lemma 1 ($\mathbb{E}_t[v_t] = \nabla f(x_t)$):

$$\mathbb{E}_t[f(x_{t+1})] \le f(x_t) - \eta\|\nabla f(x_t)\|^2 + \frac{L\eta^2}{2}\mathbb{E}_t\|v_t\|^2. \qquad \text{[Step 9]}$$

Step B: Lyapunov function. Define, for constants $c_t \ge 0$ to be chosen,

$$R_t := \mathbb{E}\big[f(x_t) + c_t\|x_t - \tilde{x}\|^2\big]. \qquad \text{[Step 10]}$$

We bound R_{t+1}. From Lemma 3 (with the chosen β) and [Step 9]:

$$\begin{aligned} R_{t+1} &= \mathbb{E}[f(x_{t+1})] + c_{t+1}\,\mathbb{E}\|x_{t+1} - \tilde{x}\|^2 \\ &\le \mathbb{E}[f(x_t)] - \eta\,\mathbb{E}\|\nabla f(x_t)\|^2 + \frac{L\eta^2}{2}\mathbb{E}\|v_t\|^2 \\ &\quad + c_{t+1}\Big[(1+\beta)\mathbb{E}\|x_t - \tilde{x}\|^2 + (1 + \tfrac{1}{\beta})\eta^2\mathbb{E}\|v_t\|^2\Big]. \qquad \text{[Step 11]} \end{aligned}$$

Step C: Insert the variance bound (Lemma 2). Let $A := \frac{L\eta^2}{2} + c_{t+1}(1 + \frac{1}{\beta})\eta^2 \ge 0$ be the coefficient of $\mathbb{E}\|v_t\|^2$. Using $\mathbb{E}\|v_t\|^2 \le 2\mathbb{E}\|\nabla f(x_t)\|^2 + 2L^2\mathbb{E}\|x_t - \tilde{x}\|^2$:

$$\begin{aligned} R_{t+1} \le \mathbb{E}[f(x_t)] &- \eta\,\mathbb{E}\|\nabla f(x_t)\|^2 + A\big(2\mathbb{E}\|\nabla f(x_t)\|^2 + 2L^2\mathbb{E}\|x_t - \tilde{x}\|^2\big) \\ &+ c_{t+1}(1+\beta)\mathbb{E}\|x_t - \tilde{x}\|^2. \qquad \text{[Step 12]} \end{aligned}$$

Collecting terms:

$$\begin{aligned} R_{t+1} \le \mathbb{E}[f(x_t)] &- (\eta - 2A)\,\mathbb{E}\|\nabla f(x_t)\|^2 \\ &+ \underbrace{\big[c_{t+1}(1+\beta) + 2AL^2\big]}_{=:\,c_t}\,\mathbb{E}\|x_t - \tilde{x}\|^2. \qquad \text{[Step 13]} \end{aligned}$$

Step D: Recursion for c_t. Define the recursion (going backward from $t = m$ to $t = 0$)

$$c_t = c_{t+1}\Big(1 + \beta + 2(1 + \tfrac{1}{\beta})\eta^2 L^2\Big) + L^3\eta^2, \qquad c_m = 0. \qquad \text{[Step 14]}$$

(Here we absorbed $2AL^2 = 2L^2\big(\frac{L\eta^2}{2} + c_{t+1}(1 + \frac{1}{\beta})\eta^2\big) = L^3\eta^2 + 2c_{t+1}(1 + \frac{1}{\beta})\eta^2 L^2$.) With $c_m = 0$ the boundary $\|x_m - \tilde{x}\|^2$ term carries no Lyapunov weight at the end of the inner loop.

Step E: Define the per-step contraction constant. Let

$$\Gamma_t := \eta - 2A = \eta - L\eta^2 - 2c_{t+1}\Big(1 + \tfrac{1}{\beta}\Big)\eta^2. \qquad \text{[Step 15]}$$

Then [Step 13] becomes the one-step descent in the Lyapunov function:

$$\Gamma_t\, \mathbb{E}\|\nabla f(x_t)\|^2 \le R_t - R_{t+1}. \qquad \text{[Step 16]}$$

We require $\Gamma_t > 0$ for all t; this is established rigorously in the Rate Derivation (Step 23–Step 27) under the stated choice of m, η, β.

Step F: Telescoping over the inner loop. Sum [Step 16] for $t = 0, \dots, m-1$. Since $\sum_t (R_t - R_{t+1}) = R_0 - R_m$ and $x_0 = \tilde{x}^s$ so $\|x_0 - \tilde{x}\|^2 = 0$ (hence $R_0 = \mathbb{E}f(\tilde{x}^s)$), and $c_m = 0$ so $R_m = \mathbb{E}f(x_m)$. Under the Option I snapshot rule $\tilde{x}^{s+1} = x_m$, we have $\mathbb{E}f(x_m) = \mathbb{E}f(\tilde{x}^{s+1})$. Therefore

$$\sum_{t=0}^{m-1} \Gamma_t\, \mathbb{E}\|\nabla f(x_t^s)\|^2 \le \mathbb{E}f(\tilde{x}^s) - \mathbb{E}f(\tilde{x}^{s+1}). \qquad \text{[Step 17]}$$

Step G: Telescoping over epochs. Let $\Gamma := \min_{0 \le t \le m-1} \Gamma_t > 0$. Then from [Step 17],

$$\Gamma \sum_{t=0}^{m-1} \mathbb{E}\|\nabla f(x_t^s)\|^2 \le \mathbb{E}f(\tilde{x}^s) - \mathbb{E}f(\tilde{x}^{s+1}). \qquad \text{[Step 18]}$$

Sum over $s = 0, \dots, S-1$ and telescope:

$$\Gamma \sum_{s=0}^{S-1} \sum_{t=0}^{m-1} \mathbb{E}\|\nabla f(x_t^s)\|^2 \le f(\tilde{x}^0) - \mathbb{E}f(\tilde{x}^S) \le f(\tilde{x}^0) - f^\star, \qquad \text{[Step 19]}$$

where the last step uses Assumption 2 ($\mathbb{E}f(\tilde{x}^S) \ge f^\star$).

Step H: Uniform sampling of output. With $T = mS$ and x_a chosen uniformly from all T inner iterates,

$$\mathbb{E}\|\nabla f(x_a)\|^2 = \frac{1}{T} \sum_{s=0}^{S-1} \sum_{t=0}^{m-1} \mathbb{E}\|\nabla f(x_t^s)\|^2. \qquad \text{[Step 20]}$$

Combining [Step 19] and [Step 20]:

$$\boxed{\mathbb{E}\|\nabla f(x_a)\|^2 \le \frac{f(\tilde{x}^0) - f^\star}{T\,\Gamma}} \qquad \text{[Step 21]}$$

Rate Derivation (With Constants)

Bounding c_t. Define $\theta := 1 + \beta + 2(1 + \frac{1}{\beta})\eta^2 L^2$ and $b := L^3\eta^2$. The recursion [Step 14] is $c_t = \theta c_{t+1} + b$, $c_m = 0$, giving

$$c_0 = b\,\frac{\theta^m - 1}{\theta - 1}. \tag{Step 22}$$

Choose $\beta = 1/m$, so $1 + \frac{1}{\beta} = 1 + m$. With $m = \lfloor n^{1/2} \rfloor \le n^{1/2}$ and $\eta = \mu_0/(Ln^{1/3})$ we have $\eta^2 L^2 = \mu_0^2/n^{2/3}$, hence

$$2(1 + \tfrac{1}{\beta})\eta^2 L^2 = 2(1+m)\,\frac{\mu_0^2}{n^{2/3}} \le 2(1 + n^{1/2})\,\frac{\mu_0^2}{n^{2/3}} \le \frac{4\mu_0^2}{n^{1/6}} \;\le\; \frac{4\mu_0^2}{1}, \tag{Step 23a}$$

where we used $1 + n^{1/2} \le 2n^{1/2}$ for $n \ge 1$ and $n^{1/2}/n^{2/3} = n^{-1/6} \le 1$. For a small constant $\mu_0 \le 1$ this term is at most $4\mu_0^2/n^{1/6} \le 4\mu_0^2$. Consequently

$$\theta \;\le\; 1 + \frac{1}{m} + \frac{4\mu_0^2}{n^{1/6}} \;\le\; 1 + \frac{1}{m} + \frac{4\mu_0^2}{m}, \tag{Step 23b}$$

where the last inequality uses $m = \lfloor n^{1/2} \rfloor \le n^{1/2} \le n^{1/6}$ for $n \ge 1$, so $1/n^{1/6} \le 1/m$. Therefore, using $(1 + a/m)^m \le e^a$,

$$\theta^m \;\le\; \Big(1 + \frac{1 + 4\mu_0^2}{m}\Big)^m \;\le\; e^{\,1+4\mu_0^2} \;=:\; C_0, \tag{Step 23}$$

a bounded constant independent of n (since $\mu_0 \le 1$ gives $C_0 \le e^5$). This corrects the earlier flawed scaling: with $m = \sqrt{n}$ (not $n^{3/2}$) the factor $2(1+m)\eta^2 L^2$ no longer grows with n.

Since $\theta - 1 = \frac{1}{m} + \frac{4\mu_0^2}{n^{1/6}} \ge \frac{1}{m}$, we obtain from [Step 22]

$$c_t \le c_0 \le b\,\frac{C_0 - 1}{\theta - 1} \le L^3\eta^2 \cdot \frac{C_0 - 1}{1/m} = L^3\eta^2 m(C_0 - 1). \tag{Step 24}$$

Plugging $\eta = \mu_0/(Ln^{1/3})$, $m \le n^{1/2}$:

$$c_t \le L^3 \cdot \frac{\mu_0^2}{L^2 n^{2/3}} \cdot n^{1/2}(C_0 - 1) = L\,\mu_0^2(C_0 - 1)\,n^{-1/6} \;\le\; L\,\mu_0^2(C_0 - 1). \tag{Step 25}$$

Bounding Γ. From [Step 15] with $\beta = 1/m$,

$$\Gamma_t = \eta - L\eta^2 - 2c_{t+1}(1+m)\eta^2. \tag{Step 26}$$

We bound the two correction terms relative to $\eta = \mu_0/(Ln^{1/3})$.

First correction.

$$L\eta^2 = \frac{\mu_0^2}{Ln^{2/3}} = \eta \cdot \frac{\mu_0}{n^{1/3}} \le \eta\,\mu_0,$$

using $n^{1/3} \ge 1$.

Second correction. Using [Step 25] ($c_{t+1} \le L\mu_0^2(C_0 - 1)$) and $1 + m \le 2n^{1/2}$,

$$2c_{t+1}(1+m)\eta^2 \le 2\,L\mu_0^2(C_0 - 1) \cdot 2n^{1/2} \cdot \frac{\mu_0^2}{L^2 n^{2/3}} = \frac{4\mu_0^4(C_0 - 1)}{L}\,n^{1/2-2/3} = \frac{4\mu_0^4(C_0 - 1)}{L}\,n^{-1/6}.$$

Dividing by $\eta = \mu_0/(Ln^{1/3})$,

$$\frac{2c_{t+1}(1+m)\eta^2}{\eta} \le 4\mu_0^3(C_0 - 1)\,n^{-1/6+1/3} = 4\mu_0^3(C_0 - 1)\,n^{1/6}.$$

This residual term grows like $n^{1/6}$, so to keep it controlled we choose μ_0 small *relative to* n: it suffices to take

$$\mu_0 \;\le\; \min\Big\{1,\; \Big(\tfrac{1}{16(e^5-1)\,n^{1/6}}\Big)^{1/3}\Big\}, \qquad \text{[Step 27a]}$$

which guarantees $4\mu_0^3(C_0-1)n^{1/6} \le 1/4$ (using $C_0 \le e^5$). Together with $L\eta^2/\eta = \mu_0/n^{1/3} \le \mu_0 \le 1/4$ (for $\mu_0 \le 1/4$), the two corrections sum to at most $\eta/2$:

$$L\eta^2 + 2c_{t+1}(1+m)\eta^2 \le \frac{\eta}{4} + \frac{\eta}{4} = \frac{\eta}{2}.$$

Hence for all t,

$$\Gamma_t \;\ge\; \eta - \frac{\eta}{2} = \frac{\eta}{2} = \frac{\mu_0}{2Ln^{1/3}} \;=:\; \Gamma \;>\; 0. \qquad \text{[Step 27]}$$

Final rate. Substitute [Step 27] into [Step 21]:

$$\mathbb{E}\|\nabla f(x_a)\|^2 \le \frac{f(\tilde{x}^0) - f^\star}{T \cdot \frac{\mu_0}{2Ln^{1/3}}} = \frac{2Ln^{1/3}\big(f(\tilde{x}^0) - f^\star\big)}{\mu_0\, T}. \qquad \text{[Step 28]}$$

Thus

$$\boxed{\mathbb{E}\|\nabla f(x_a)\|^2 = O\left(\frac{Ln^{1/3}\,(f(\tilde{x}^0) - f^\star)}{T}\right).}$$

The bound is $O(n^{1/3}/T)$ in T; the factor $n^{1/3}$ does not vanish as $T \to \infty$, so the rate is *not* the n-independent $O(1/T)$ erroneously claimed earlier.

Oracle complexity. To achieve $\mathbb{E}\|\nabla f(x_a)\|^2 \le \epsilon$ we need, from [Step 28],

$$T = O\Big(\frac{Ln^{1/3}}{\epsilon}\Big)$$

inner iterations. The number of epochs is $S = T/m = O\big(Ln^{1/3}/(\epsilon\, n^{1/2})\big) = O\big(L/(\epsilon\, n^{1/6})\big)$, each costing n component gradients for the full-gradient snapshot. The total gradient cost is therefore

$$O\big(n \cdot S + T\big) = O\Big(\frac{n^{5/6}}{\epsilon} + \frac{n^{1/3}}{\epsilon}\Big) = O\Big(n + \frac{n^{5/6}}{\epsilon}\Big),$$

where the n term covers at least one full snapshot pass. This improves on full-batch GD's $O(n/\epsilon)$ whenever $n^{5/6}/\epsilon \lesssim n/\epsilon$, i.e. always for $n \ge 1$, and remains far better than SGD's $O(1/\epsilon^2)$ for the regimes of interest.

Special Cases

- $n = 1$ **(single function).** The estimator collapses to $v_t = \nabla f(x_t)$ and SVRG reduces to gradient descent; [Step 9] becomes the standard descent inequality $\mathbb{E}f(x_{t+1}) \le f(x_t) - (\eta - \frac{L\eta^2}{2})\|\nabla f(x_t)\|^2$, giving $\frac{1}{T}\sum \|\nabla f(x_t)\|^2 = O(1/T)$ with $\eta \le 1/L$. With $n = 1$ the factor $n^{1/3} = 1$ and the general bound [Step 28] reduces to exactly this rate, consistent with the dry-run example.

- **Convex case.** If in addition f is convex, one can show $\mathbb{E}[f(x_a) - f^\star] = O(1/T)$ with the standard SVRG analysis (linear convergence under strong convexity), a stronger guarantee than the stationarity bound here.

- **Large m.** If m is taken too large (e.g. $m \sim n^{3/2}$ with the same η), the factor $2(1+m)\eta^2 L^2$ entering θ no longer stays $O(1/m)$, so θ^m ceases to be bounded, c_t blows up, and Γ_t can become negative. This is precisely why the corrected choice $m = O(n^{1/2})$ and $\eta = O(1/(Ln^{1/3}))$ are jointly required to keep $\Gamma \geq \eta/2 > 0$ (Step 23–Step 27).